The p^+–integrated four–kernel row

Is it right, where the rapidity logarithm is, and what comes next

$\kappa \equiv k\cdot(y'-z),\ \ \xi \equiv p^+/k^+,\ \ \lambda \equiv \Lambda/k^+,\ \ \Xi \equiv (\vee - k^+)/k^+,\ \ \ell_k \equiv \log(k^+/\Lambda) = -\log\lambda$

1. Verdict

> **All three p^+ integrals are correct** — structures, signs, arguments, and the $e^{ik\cdot(y'-z)}$ prefactor on the third line. Each was checked against numerical quadrature to 10^{-29} relative.
>
> **One thing to check: the overall $1/k^+$.** See §2.

The bracket before integration is $\frac{\delta_{k'm}\delta_{ij}}{p^++k^+} - \frac{\delta_{jm}\delta_{ik'}}{p^+} - \frac{\delta_{im}\delta_{jk'}}{k^+}$, which on the four kernels collapses to

$$\frac{1}{k^+}\left[\frac{A}{1+\xi} - \frac{B}{\xi} - C\right], \qquad \begin{aligned} A &= (y'-z)\cdot(x'-y')\ (y-w)\cdot(x-z) \\ B &= (y'-z)\cdot(x-z)\ (x'-y')\cdot(y-w) \\ C &= (y'-z)\cdot(y-w)\ (x'-y')\cdot(x-z) \end{aligned}$$

(all divided by $(y'-z)^2(x'-y')^2(y-w)^2(x-z)^2$), with the phase $e^{-i\kappa\xi}$ from $e^{-ip\cdot(y'-z)}$ at $p_\perp = \xi k$. Then $\frac{1}{k^+}\int dp^+ = \int d\xi$ and

structure	$\int_\lambda^\Xi d\xi\,(\cdot)\,e^{-i\kappa\xi}$	your line	check
$-C$	$\dfrac{e^{-i\kappa\Xi} - e^{-i\kappa\lambda}}{i\kappa}$	1st	4.8×10^{-29}
$-B/\xi$	$-\big[\mathrm{Ei}(-i\kappa\Xi) - \mathrm{Ei}(-i\kappa\lambda)\big]$	2nd	1.3×10^{-30}
$+A/(1+\xi)$	$e^{i\kappa}\big[\mathrm{Ei}(-i\kappa(1{+}\Xi)) - \mathrm{Ei}(-i\kappa(1{+}\lambda))\big]$	3rd	5.3×10^{-31}

Every one of these is exactly what you wrote, including the shift $1+\Xi = \vee/k^+$ inside the last Ei and the $e^{ik\cdot(y'-z)}$ that the shift $u = 1+\xi$ generates.

2. The one thing to check: the overall $1/k^+$

> In your first line the denominator is $i\,k\cdot(y'-z)$ — **not** $i\,k\cdot(y'-z)/k^+$. That missing k^+ is the fingerprint: it was already spent by $dp^+ = k^+d\xi$ (equivalently, you integrated $d\xi$ directly). So the $\frac{1}{k^+}$ that you factored out of $\frac{1}{p^++k^+} = \frac{1}{k^+(1+\xi)}$ has **already been cancelled** by the measure.
>
> Concretely: $\frac{1}{k^+}\int dp^+\big[\cdots\big] = \int d\xi\big[\cdots\big] = \{\text{your braces}\}$, with *no* $1/k^+$ left over.
>
> So the $\frac{1}{k^+}$ standing in front of your braces is legitimate **only if** it was in the prefactor *before* the bracket — e.g. from the $\frac{1}{2k^+}$ of the gluon phase space. If it is the one you pulled out of the bracket, the answer is too small by k^+. Trace it back to the pre–integration expression; that is a one–line check.

3. Extracting the rapidity logarithm

Two expansions do all the work. For $\varepsilon \to 0^+$,

$$\mathrm{Ei}(-i\kappa\varepsilon) = \gamma_E + \log\big(i\kappa\varepsilon\big) + O(\kappa\varepsilon), \qquad \mathrm{Ei}(-i\kappa X) \xrightarrow[X\to\infty]{} -i\pi\,\mathrm{sgn}\,\kappa.$$

The second is worth pausing on: Ei at large imaginary argument does **not** go to zero, it goes to the constant $\mp i\pi$ ($|\mathrm{Ei}(-i\,10^6)| = 3.141592$). Dropping the upper limit is therefore wrong by $i\pi$.

Line by line, as $\Lambda \to 0$ and $\vee \to \infty$

Third line, $A/(1+\xi)$: no rapidity log. The denominator $1+\xi$ is regular at $\xi = 0$, so $\mathrm{Ei}(-i\kappa(1+\lambda)) \to \mathrm{Ei}(-i\kappa)$ smoothly:

$$\longrightarrow\ \ e^{i\kappa}\big[-i\pi\,\mathrm{sgn}\,\kappa - \mathrm{Ei}(-i\kappa)\big].$$

First line, $-C$: no rapidity log. $e^{-i\kappa\lambda}\to 1$, and the Ξ piece is bounded and oscillatory (Riemann–Lebesgue kills it under the transverse integration):

$$\longrightarrow\ \frac{i}{\kappa}\ +\ (\text{oscillatory}).$$

This also retires the worry I raised last time that this term looked linear in $\vee$. It is not — the phase saves it.

Second line, $-B/\xi$: the sole rapidity divergence. In the double limit the bracket collapses to a single logarithm,

$$-\big[\mathrm{Ei}(-i\kappa\Xi)-\mathrm{Ei}(-i\kappa\lambda)\big]\ \longrightarrow\ \gamma_E+\log\big(i\kappa\lambda\big)\ =\ \boxed{-\,\ell_k\ +\ \gamma_E+\log\big(i\,k\cdot(y'-z)\big)}$$

verified to 10^{-9} for both signs of κ (the log is the principal branch, and it carries the $\pm i\pi/2$ automatically). **The coefficient of ℓ_k is -1, and it sits on the structure B alone.**

4. What multiplies ℓ_k is the BK kernel

Strip the ℓ_k off and look at what it multiplies:

$$-\,\ell_k\ \underbrace{\frac{(x-z)\cdot(y'-z)}{(x-z)^2\,(y'-z)^2}}_{\mathcal{K}(x,y';z)}\ \times\ \underbrace{\frac{(x'-y')\cdot(y-w)}{(x'-y')^2\,(y-w)^2}}_{\text{no }z\text{: LO structure}}$$

The first factor is *exactly* the real–emission piece of the Balitsky–Kovchegov / JIMWLK dipole kernel, with z the emission point and x,y' the two legs. That is the strongest available check that the p^+ integration is right: the rapidity divergence had to come out proportional to the evolution kernel, and it does.

And this is where last note's transverse IR log goes

In the previous note I found a logarithmic divergence of the z–integration at large $|z|$, of coefficient $-\pi\delta^{mj}$, and observed that it is tied to $\xi\to 0$. The full BK kernel is

$$\mathcal{M}(x,y';z)=\frac{(x-y')^2}{(x-z)^2(y'-z)^2}=\frac{1}{(x-z)^2}+\frac{1}{(y'-z)^2}-2\,\mathcal{K}(x,y';z),$$

and the two $1/(\cdot)^2$ terms are the **virtual** (self–energy / same–side) contributions of the same order, which are *not* in the row you wrote. Their large–$|z|$ angular averages are what cancel the divergence:

ρ	$\rho^2\langle\frac{1}{(x-z)^2}\rangle$	$\rho^2\langle\frac{1}{(y'-z)^2}\rangle$	$\rho^2\langle -2\mathcal{K}\rangle$	$\rho^2\langle\mathcal{M}\rangle$
10^1	1.001302	1.008980	-1.993815	0.016466
10^2	1.000013	1.000089	-1.999938	0.000164
10^3	1.000000	1.000001	-1.999999	0.000002
10^4	1.000000	1.000000	-2.000000	0.000000

$1+1-2=0$: the full kernel falls as ρ^{-4}, not ρ^{-2}, and the z–integral converges. **So the transverse infrared logarithm is not a disease of your calculation — it is telling you that the virtual diagrams are still missing.**

5. What to do next

1. **Settle the** $1/k^+$ (§2) against the pre–integration expression.
2. **Take $\vee\to\infty$ honestly.** Keep the $-i\pi\,\mathrm{sgn}\kappa$ from $\mathrm{Ei}(-i\kappa\Xi)$ — it is a finite constant, not zero, and it contributes to the imaginary (absorptive) part.
3. **Write the three brackets in their limiting forms** (§3). You then have a clean split: one term $\propto\ell_k$, everything else ℓ_k–independent.
4. **Collect the virtual partners** so that $-2\mathcal{K}\to\mathcal{M}$. Until you do, the z–integral is IR divergent and no separation into “finite” pieces is meaningful.

5. **Reduce the colour structure.** $f^{c'd'a}$ contracted with $[UU\rho - UU\rho]\,[UUU\rho\rho - U\rho\rho]$ has to be brought to dipole / quadrupole operators before the ℓ_k coefficient can be compared with $\mathcal{H}_{\text{JIMWLK}}$. The check to aim for is

$$\left.\frac{d^3 N_{\text{NLO}}}{d^3k}\right|_{\ell_k} = -\,\ell_k\ \mathcal{H}_{\text{JIMWLK}} \otimes \frac{d^3 N_{\text{LO}}}{d^3k}.$$

6. **Absorb, don't cancel.** Once (5) holds, define the target weight functional at rapidity $Y = \ell_k$ and subtract; what remains is the genuine NLO impact factor.
7. **Then** do the transverse integrals with the master $B^{mj}(Q, c)$ of the previous note — but note the phase is now gone (you integrated p^+ first), so the roles of Λ and Q are swapped: Λ is what cuts the large–separation region, and the leftover $\log(i\,k \cdot (y' - z))$ from §3 is what turns the single ℓ_k into the double log ℓ_k^2 found earlier.

`zint/src/pplus_check.py`, `zint/src/pplus_limits.py` — every number above, written to `zint/RESULTS_pplus.txt`.